Matrices (continued)

Key terms **KSBs: K13, S11, S52, S53, B1**

identity matrix, inverse matrix, Gauss-Jordan elimination, row operations, systems of linear equations, covariance matrix

Definition 10.1 - Identity Matrix

The **identity matrix** is a square matrix I such that

$$[I]_{ij} = \begin{cases} 1 & i=j \\ 0 & \text{otherwise} \end{cases}$$

For $n = 3$, the matrix is

$$I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

The identity matrix is the equivalent of the number 1 in multiplication - if you multiply any matrix by the identity, it will not change. That is, for any matrix A, $IA = AI = A$.

Example 10.1.

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \qquad AI = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

10.1 Inverse matrices

We have seen that matrix multiplication is defined for matrices of the right size. There is no equivalent definition of division for matrices. However, for certain square matrices it may be possible to compute the inverse matrix which can be multiplied by the matrix to achieve the same effect.

For matrices, the multiplicative identity is the identity matrix, so we can define a multiplicative inverse with respect to matrix multiplication, and for a matrix A its inverse is denoted A^{-1}.

Definition 10.2 - Inverse Matrix

Let A be a non-zero $n \times n$ matrix. Then, if there exists an $n \times n$ matrix A^{-1} such that $A^{-1}A = I_n$ we say that A^{-1} is the inverse of the matrix A.

Matrix icon by Adan24 from the Noun Project

Example 10.2. *Given the matrix* $A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ *then its inverse is* $A^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$. *Indeed,*

$$AA^{-1} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

We could also show this by calculating $A^{-1}A$:

$$\begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

In order for a matrix to be invertible, it needs:

- to be a square matrix
- to have a non-zero determinant.

Note

- Not all square matrices have an inverse
- If a matrix does not have an inverse, it has a determinant of zero and is said to be singular.
- If a matrix has an inverse, it has a non-zero determinant and it is said to be invertible.
- The inverse of a square matrix is unique.
- Even though multiplying matrices the other way round doesn't generally give the same result, we can always say that $A^{-1}A = AA^{-1} = I_n$.

Definition 10.3 - Inverse of a 2-by-2 Matrix

For a two-by-two matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the inverse is given by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

That is, the matrix with the entries flipped along the main diagonal and the diagonal entries made negative, divided by the determinant of the matrix.

Note

Note that the determinant must be non-zero for this to be defined.

Example 10.3. *The inverse of this 2-by-2 matrix can be found using the formula given in the definition:*

$$\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}^{-1} = \frac{1}{(1\times 2)-(1\times 1)}\begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}.$$

Exercise 10.1. *Calculate the inverses of the following matrices if they exist:*

(i) $A = \begin{pmatrix} 1 & 0 \\ 3 & 2 \end{pmatrix}$

(ii) $B = \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix}$

(iii) $C = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

Let A and B be two invertible $n \times n$ matrices. Then the following hold:

- AB is also invertible and $(AB)^{-1} = B^{-1}A^{-1}$.
- $det(A^{-1}) = \frac{1}{det(A)}$
- A^{T} is also invertible and $(A^{\mathsf{T}})^{-1} = (A^{-1})^{\mathsf{T}}$

Calculating the inverse of a matrix (called inverting the matrix) can be done using a variety of methods. For 2-by-2 matrices, the formula above gives the inverse. For larger matrices, we can use Gauss-Jordan elimination.

10.2 Gauss-Jordan Elimination

In order to perform Gauss-Jordan Elimination, we need to use **row operations**.

Definition 10.4 - Row Operations

The three **elementary row operations** that can be applied to a matrix are:

- **Type I**: swap any two rows of the matrix.
- **Type II**: multiply one row by a non-zero ~~scalar~~ scalar
- **Type III**: replace a single row by itself plus a multiple of another row

Example 10.4.

$$\begin{pmatrix} 3 & 1 & -2 & 1 \\ 1 & -1 & 2 & 3 \\ 2 & -3 & 7 & 4 \end{pmatrix}$$

$$\begin{matrix} R_2 - 3R_1 \\ R_3 - 2R_1 \end{matrix} \rightarrow \begin{pmatrix} 1 & -1 & 2 & 3 \\ 0 & 4 & -8 & -8 \\ 0 & -1 & 3 & -2 \end{pmatrix}$$

$$R_1 \leftrightarrow R_2 \longrightarrow \begin{pmatrix} 1 & -1 & 2 & 3 \\ 3 & 1 & -2 & 1 \\ 2 & -3 & 7 & 4 \end{pmatrix}$$

$$\begin{matrix} R_3 \rightarrow 2 \times R_3 \\ (R_3 \times 2) \end{matrix} \quad \begin{pmatrix} 1 & -1 & 2 & 3 \\ 0 & 4 & -8 & -8 \\ 0 & -2 & 6 & -4 \end{pmatrix}$$

Exercise 10.2. *Let A be the matrix:*

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

Perform the following row operations on the matrix A (each time starting again with the original matrix).

(i) $R_1 \leftrightarrow R_2$

(ii) $R_2 \rightarrow 3R_2$

(iii) $R_3 \rightarrow R_3 + 2R_2$

These operations will change the numbers in the matrix, and will also change the properties of the matrix (including its determinant), but if they are applied consistently, they can be used to transform the matrix in a useful way.

Definition 10.5 - Gauss-Jordan Elimination

The process of **Gauss-Jordan Elimination** involves applying row operations, one at a time, to a matrix, until the matrix has been completely reduced to the identity matrix.

Example 10.5.

$$A = \begin{pmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 1 & 4 & 4 \end{pmatrix}$$

Performing row reduction:

$$\begin{pmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 1 & 4 & 4 \end{pmatrix}$$

$$\begin{matrix} R_2 - 2R_1 \\ R_3 - R_1 \end{matrix} \longrightarrow \begin{pmatrix} 1 & 0 & 2 \\ 0 & -1 & -1 \\ 0 & 4 & 2 \end{pmatrix}$$

$$-R_2 \longrightarrow \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 4 & 2 \end{pmatrix}$$

$R_2 \to (-1) \times R_2$

$$R_3 - 4R_2 \to \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & -2 \end{pmatrix}$$

$$-\tfrac{1}{2} \times R_3 \to \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{matrix} R_1 - 2R_3 \\ R_2 - R_3 \end{matrix} \to \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Exercise 10.3. *For each of the matrices below, perform row operations to reduce it to the identity matrix.*

$$(a) \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix} \qquad (b) \begin{pmatrix} 2 & 3 & 0 \\ 1 & -2 & -1 \\ 2 & 0 & -1 \end{pmatrix} \qquad (c) \begin{pmatrix} 1 & 1 & 1 \\ 2 & 3 & 7 \\ 1 & 3 & -2 \end{pmatrix}$$

10.3 Inverting Matrices using Gauss-Jordan

To use Gauss-Jordan elimination to invert a matrix, we write a copy of the matrix next to a copy of the identity matrix (using a vertical line to separate the two). We call this an **augmented matrix**.

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 2 & -1 & 3 & 0 & 1 & 0 \\ 1 & 4 & 4 & 0 & 0 & 1 \end{array}\right)$$

Then, we perform row operations to the whole matrix, until the left-hand side has become the identity matrix. The same series of steps, applied to the identity matrix, will convert it into the inverse of our original matrix.

Example 10.6. *Using the same steps from Example 10.5, we can find the inverse of the matrix.*

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 2 & -1 & 3 & 0 & 1 & 0 \\ 1 & 4 & 4 & 0 & 0 & 1 \end{array}\right)$$

$$\begin{array}{c} R_2 - 2R_1 \\ R_3 - R_1 \end{array} \longrightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & -1 & -1 & -2 & 1 & 0 \\ 0 & 4 & 2 & -1 & 0 & 1 \end{array}\right)$$

$$-R_2 \longrightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 2 & -1 & 0 \\ 0 & 4 & 2 & -1 & 0 & 1 \end{array}\right)$$

$$R_3 - 4R_2 \rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 2 & -1 & 0 \\ 0 & 0 & -2 & -9 & 4 & 1 \end{array}\right)$$

$$-\tfrac{1}{2}R_3 \rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 2 & -1 & 0 \\ 0 & 0 & 1 & 9/2 & -2 & -\tfrac{1}{2} \end{array}\right)$$

$$\begin{array}{c} R_1 - 2R_3 \\ R_2 - R_3 \end{array} \rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -8 & 4 & 1 \\ 0 & 1 & 0 & -5/2 & 1 & 1/2 \\ 0 & 0 & 1 & 9/2 & -2 & -\tfrac{1}{2} \end{array}\right)$$

Hence the inverse matrix is given by

$$A^{-1} = \begin{pmatrix} -8 & 4 & 1 \\ -2.5 & 1 & 0.5 \\ 4.5 & -2 & -0.5 \end{pmatrix}$$

$$AA^{-1} = \begin{pmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 1 & 4 & 4 \end{pmatrix}\begin{pmatrix} -8 & 4 & 1 \\ -2.5 & 1 & 0.5 \\ 4.5 & -2 & -0.5 \end{pmatrix} =$$

Exercise 10.4. *Use Gauss-Jordan elimination to calculate the inverse of the following matrices.*

(a) $\begin{pmatrix} -4 & 2 \\ 3 & 4 \end{pmatrix}$

(b) $\begin{pmatrix} -4 & 2 \\ -1 & -3 \end{pmatrix}$

(c) $\begin{pmatrix} -4 & -4 & -2 \\ 3 & 0 & 4 \\ 1 & 0 & 0 \end{pmatrix}$

(d) $\begin{pmatrix} 4 & 0 & -4 \\ 4 & -1 & 1 \\ 3 & 1 & 2 \end{pmatrix}$

10.4 Systems of Linear Equations

Definition 10.6 - Linear Equation

A linear equation is an equation that involves a set of variables multiplied by scalar coefficients where the sum of these is equal to some constant scalar value. Generally, if $x_1, x_2, \ldots, x_n$ are variables, $a_1, a_2, \ldots, a_n$ are coefficients and b is a constant value then a linear equation can take the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

Example 10.7. *Linear equations include:*

$$3x+2y=5 \qquad x=2 \qquad x+y+z=2$$

The following are not linear equations:

$$x^2=4 \qquad e^x=y \qquad xy=5$$

Linear equations often describe systems in the real world. Frequently, more complex systems can be described using collections of linear equations involving the same set of variables which all need to be true at the same time, which we call a system of linear equations.

Example 10.8. *A car showroom sells three types of vehicles: saloons, hatchbacks and people carriers. On Thursday, the showroom sold* 3 *saloons,* no *hatchbacks and* 2 *people carriers, and made* £174k. *On Friday, they sold* 2 *saloons and* no *hatchbacks, and* 2 *people carriers were returned for a full refund, and the overall income was* −£24k. *On Saturday, they sold* 1 *hatchback and* 1 *people carrier, and made* £57k.

Let S represent the profit on a saloon car, H represent the profit on a hatchback, and P represent the profit on a people carrier. Then we can write an equation for each day:

$$3S+2P=174$$
$$2S-2P=-24$$
$$H+P=57$$

By solving these equations, we could find a set of values for S, H and P *which would represent the cost of each vehicle type in thousands of pounds.*

Exercise 10.5. *Write down a system of linear equations corresponding to the situations given.*

(a) *Tickets for a play are sold in three areas: standing tickets, seated tickets and balcony tickets. On Thursday night, the venue sold 36 standing tickets, 12 seated tickets and 9 balcony tickets, and made \$612. On Friday, they made \$840 and sold 40 standing tickets, 24 seated tickets and 10 balcony tickets. On Saturday, they sold 2 more of each type of ticket than they did on Friday, and made \$906. How much does each ticket type cost?*

(b) *At a dog show the dogs were each given a bowl of food, and the audience members and photographers present were given two bowls of food each. Photographers were assigned two seats in the auditorium (so they'd have space for all their kit), audience members 1 seat each, and dogs were asked to share, two to a seat. Altogether, 188 seats were used, 334 bowls of food were eaten, and 535 legs were present (including the legs on each photographer's tripod). All the humans present had two legs. How many dogs, audience members and photographers were there?*

(c) *You are playing a quiz game. Getting a question correct gives you 100 points, and getting an answer wrong gives you a penalty of -25 points. You have answered 40 questions, and had a final score of 2250 points. How many questions did you get right and wrong?*

10.5 Representing systems of equations using matrices

Matrices can be used to represent systems of linear equations, and manipulating the matrix allows us to find solutions to the system.

Example 10.9. *For the system of equations given in Example 10.8, we can write down a* ***matrix of coefficients*** *representing the relationship between the variables.*

$$3S + 2P = 174$$
$$2S - 2P = -24$$
$$H + P = 57$$

$$A = \begin{array}{c} \begin{matrix} S & H & P \end{matrix} \\ \begin{pmatrix} 3 & 0 & 2 \\ 2 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix} \end{array}$$

We can write the whole system as:

$$\begin{pmatrix} 3 & 0 & 2 \\ 2 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} S \\ H \\ P \end{pmatrix} = \begin{pmatrix} 174 \\ -24 \\ 57 \end{pmatrix}$$

Here, the matrix of coefficients A is multiplied by a vector containing the variables S, H & P *which we often denote* $\underline{x}$ *and the result is equal to the* vector of constants *corresponding to the right-hand sides of the equations, which we can denote* $\underline{b}$

If we multiply the left hand side together, we obtain the vector:

$$\begin{pmatrix} 3 & 0 & 2 \\ 2 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} S \\ H \\ P \end{pmatrix} = \begin{pmatrix} 3S + 2P \\ 2S - 2P \\ H + P \end{pmatrix}$$

If this is set equal to the vector of constants, we obtain the same system of equations:

$$\begin{pmatrix} 3S + 2P \\ 2S - 2P \\ H + P \end{pmatrix} = \begin{pmatrix} 174 \\ -24 \\ 57 \end{pmatrix}$$

These two vectors are only equal if each pair of entries in corresponding positions are equal.

In this way we can write a system of linear equations as a matrix equation

$$A\underline{x} = \underline{b}$$

10.5.1 Solving systems of equations using inverse matrices

If we had an algebraic equation like $2x = 6$ we could solve this by dividing both sides by 2 to get $x = 3$.

In the case of matrices, recall that we cannot divide, but we can use inverse matrices to get the same result.

If $A\mathbf{x} = \mathbf{b}$, then we can multiply on the left of both sides by the inverse matrix A^{-1} to get:

$$\underbrace{A^{-1}A}\underline{x} = A^{-1}\underline{b}$$

Recall that $A^{-1}A$ is just the identity matrix and that multiplying something by the identity matrix has no effect - so this is effectively $\underline{x} = A^{-1}\underline{b}$

So if we can find the inverse of the matrix of coefficients, we can multiply this by the constant vector to get the Solutions to the system.

Example 10.10. *From Example 10.8, we had the coefficient matrix* $A = \begin{pmatrix} 3 & 0 & 2 \\ 2 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix}$

We can find the inverse of this matrix using Gauss-Jordan elimination:

$$\left(\begin{array}{ccc|ccc} 3 & 0 & 2 & 1 & 0 & 0 \\ 2 & 0 & -2 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right)$$

$$R_1 \to R_1 + R_2 \quad \cancel{R_2 + R_1} \to \left(\begin{array}{ccc|ccc} 5 & 0 & 0 & 1 & 1 & 0 \\ 2 & 0 & -2 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right)$$

$$R_1 \div 5 \to \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{5} & \frac{1}{5} & 0 \\ 2 & 0 & -2 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right)$$

$$R_2 - 2R_1 \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{5} & \frac{1}{5} & 0 \\ 0 & 0 & -2 & -\frac{2}{5} & \frac{3}{5} & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right)$$

$$R_2 \div -2 \longrightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{5} & \frac{1}{5} & 0 \\ 0 & 0 & 1 & \frac{1}{5} & -\frac{3}{10} & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right)$$

$$R_2 \leftrightarrow R_3 \longrightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{5} & \frac{1}{5} & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & \frac{1}{5} & -\frac{3}{10} & 0 \end{array}\right)$$

$$R_2 \to R_2 - R_3 \longrightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{5} & \frac{1}{5} & 0 \\ 0 & 1 & 0 & -\frac{1}{5} & \frac{3}{10} & 1 \\ 0 & 0 & 1 & \frac{1}{5} & -\frac{3}{10} & 0 \end{array}\right)$$

This means the inverse of the matrix A is

$$A^{-1} = \begin{pmatrix} \frac{1}{5} & \frac{1}{5} & 0 \\ -\frac{1}{5} & \frac{3}{10} & 1 \\ \frac{1}{5} & -\frac{3}{10} & 0 \end{pmatrix}$$

If we multiply this by the vector of constants, we get:

$$A^{-1}\underline{b} = \begin{pmatrix} \frac{1}{5} & \frac{1}{5} & 0 \\ -\frac{1}{5} & \frac{3}{10} & 1 \\ \frac{1}{5} & -\frac{3}{10} & 0 \end{pmatrix}\begin{pmatrix} 174 \\ -24 \\ 57 \end{pmatrix} = \begin{pmatrix} \frac{1}{5}\times 174 + \frac{1}{5}\times -24 \\ -\frac{1}{5}\times 174 + \frac{3}{10}\times -24 + 57 \\ \frac{1}{5}\times 174 - \frac{3}{10}\times -24 \end{pmatrix} = \begin{pmatrix} 30 \\ 15 \\ 42 \end{pmatrix}$$

This means the cost of a saloon car is £30k, *a hatchback is* £15k *and a people carrier is* £42k.

Exercise 10.6. *By finding the inverse of the matrix of coefficients, find solutions to the systems of equations:*

$$(a)\ \begin{matrix} -4x_1 + 2x_2 = -22 \\ 3x_1 + 4x_2 = 11 \end{matrix} \qquad (b)\ \begin{matrix} -4x_1 + 2x_2 = 6 \\ -x_1 - 3x_2 = -2 \end{matrix} \qquad (c)\ \begin{matrix} -4x_1 - 4x_2 - 2x_3 = 16 \\ 3x_1 + 4x_3 = -8 \\ x_1 = 0 \end{matrix}$$

10.5.2 Solving a system using Gauss-Jordan

It is also possible to directly solve a system of linear equations using Gauss-Jordan elimination.

In order to do this, we form an augmented matrix consisting of the matrix of coefficients and the constant vector, and perform row operations until the left hand side is the identity matrix.

Example 10.11. *Again using the data from Example 10.8, we can form the augmented matrix:*

$$(A|\underline{b}) = \left(\begin{array}{ccc|c} 3 & 0 & 2 & 174 \\ 2 & 0 & -2 & -24 \\ 0 & 1 & 1 & 57 \end{array}\right)$$

We now perform Gauss-Jordan elimination to make the left side equal to the identity matrix:

$$\left(\begin{array}{ccc|c} 3 & 0 & 2 & 174 \\ 2 & 0 & -2 & -24 \\ 0 & 1 & 1 & 57 \end{array}\right)$$

$$R_1 \to R_1 + R_2 \quad \to \left(\begin{array}{ccc|c} 5 & 0 & 0 & 150 \\ 2 & 0 & -2 & -24 \\ 0 & 1 & 1 & 57 \end{array}\right)$$

$$R_1 \div 5 \quad \to \left(\begin{array}{ccc|c} 1 & 0 & 0 & 30 \\ 2 & 0 & -2 & -24 \\ 0 & 1 & 1 & 57 \end{array}\right)$$

$$\xrightarrow{R_2 - (2\times R_1)} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 30 \\ 0 & 0 & -2 & -84 \\ 0 & 1 & 1 & 57 \end{array}\right)$$

$$\xrightarrow{R_2 \div -2} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 30 \\ 0 & 0 & 1 & 42 \\ 0 & 1 & 1 & 57 \end{array}\right)$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 30 \\ 0 & 1 & 1 & 57 \\ 0 & 0 & 1 & 42 \end{array}\right)$$

$$R_2 - R_3 \left(\begin{array}{ccc|c} 1 & 0 & 0 & 30 \\ 0 & 1 & 0 & 15 \\ 0 & 0 & 1 & 42 \end{array}\right)$$

This is the same solution as before.

Exercise 10.7. *Using Gauss-Jordan elimination, find solutions to these systems of equations.*

(a) $\begin{aligned} -x_1 + 3x_2 &= -2 \\ -2x_1 + x_2 &= 1 \end{aligned}$ (b) $\begin{aligned} 3x_1 + x_2 + 2x_3 &= 11 \\ 4x_1 - 4x_3 &= -4 \\ 4x_1 - 2x_2 + x_3 &= 13 \end{aligned}$ (c) $\begin{aligned} -x_1 - 5x_2 - 2x_3 &= -17 \\ 2x_1 - 2x_2 - 3x_3 &= -14 \\ 3x_1 - x_2 + 4x_3 &= -13 \end{aligned}$

10.6 Other uses of linear algebra in data science

10.6.1 Covariance matrix

Definition 10.7 - Covariance matrix

For a collection of datasets, the **covariance matrix** is a matrix containing the variance and covariance between the various datasets.

For two variables X and Y, the covariance matrix is given by: $\begin{pmatrix} var(X) & cov(X,Y) \\ cov(Y,X) & var(Y) \end{pmatrix}$

For three variables X, Y and Z, it is $\begin{pmatrix} var(X) & cov(X,Y) & cov(X,Z) \\ cov(Y,X) & var(Y) & cov(Y,Z) \\ cov(Z,X) & cov(Z,Y) & var(Z) \end{pmatrix}$

Recall that $cov(X,Y) = cov(Y,X)$ so this matrix is equal to its own transpose.

Example 10.12. *Let X and Y be the two datasets:*

$$X = (4, 4, 8, 12) \qquad Y = (2, 5, 10, 9)$$

First the means are $\bar{x} = 7$, $\bar{y} = 6.5$

Then the variance of X is given by:

$$var(X) = \frac{1}{n}\sum(x-\bar{x})^2 = \frac{(4-7)^2+(4-7)^2+(8-7)^2+(12-7)^2}{4} = \frac{36}{4} = 9$$

The variance of Y is given by:

$$var(Y) = \frac{1}{n}\left(\sum(y-\bar{y})^2\right) = \cdots = \frac{45}{4} = 11.25$$

The covariance of X and Y is given by:

$$cov(X,Y) = \frac{\sum(x-\bar{x})(y-\bar{y})}{n} = \frac{16.5+10.5-1.5+2.5}{4} = \frac{28}{4} = 7$$

So the covariance matrix is: $\begin{pmatrix} 9 & 7 \\ 7 & 11.25 \end{pmatrix}$

This matrix tells us that X has alower variance than Y *and that the covariance of the two sets is* positive *which means they are* positively correlated.

Exercise 10.8.

(a) Using the dataset $Z = (0, 6, 12, 6)$ calculate the covariance matrix for X, Y and Z where X and Y are the sets given in the previous example.

(b) A covariance matrix for three variables is given by:

$$\begin{pmatrix} 4 & -2 & 0 \\ -2 & 3 & 1 \\ 0 & 1 & 5 \end{pmatrix}$$

Describe what the matrix tells you about the relationships between the datasets.

(c) Calculate the covariance matrix of the datasets $X = (2, 4, 6, 8, 10)$ and $Y = (1, 3, 5, 7, 9)$.

At this stage, a covariance matrix is merely a useful way to summarise this data, but in more advanced data science contexts they are used as true matrices and can be manipulated, inverted and multiplied together like any other matrix. They have applications in principal component analysis, image recognition and signal processing.

10.7 Linear Regression using Inverse Matrices

Earlier, we saw how to calculate the regression line for a two-variable dataset, and to calculate the slope and y-intercept of the line manually. For larger datasets, this becomes more complicated, and a matrix can be used to perform least squares regression.

While this is beyond the scope of this course, the formula for calculating the slope of the regression line (given a matrix of input variable values X and a vector of output variable values y) can be written:

$$b = (X^T \cdot X)^{-1} \cdot X^T \cdot y$$